NanoGT Match Report: Crigler-Najjar syndrome type I

Disease: Crigler-Najjar syndrome type I (ORPHA:1060)
Primary gene: UGT1A1
Gene CDS: 1599 bp
Inheritance: Autosomal recessive
Target tissues: liver

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	CPCB-RPE1	AAV8	6.7/10	Medium	phase2/3
2	SRP-9001	AAV9	6.7/10	Medium	approved
3	RGX-121	AAV9	6.5/10	Medium	phase3
4	DTX201	AAV8	6.3/10	Medium	phase2
5	DTX301	AAV8	6.3/10	Medium	phase2

Match #1: CPCB-RPE1

Precedent disease: Achromatopsia
Vector: AAV8
Tissue target: retina/photoreceptor
Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	1.0	2.0
Protein class	1.5	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.7	1.0

Rationale

- Gene CDS 1599bp / cargo 4700bp (34% utilized)
- Vector tropism overlaps liver, but precedent target is retina/photoreceptor
- Both membrane proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: phase2/3

Match #2: SRP-9001

Precedent disease: Duchenne muscular dystrophy
Vector: AAV9
Tissue target: muscle
Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	1.0	2.0
Protein class	1.5	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1599bp / cargo 4700bp (34% utilized)
- Vector tropism overlaps liver, but precedent target is muscle
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: approved

Match #3: RGX-121

Precedent disease: Mucopolysaccharidosis type II
Vector: AAV9
Tissue target: CNS/liver
Composite score: 6.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 1599bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement

- Unknown pathway — neutral score
- Approval status: phase3

Match #4: DTX201

Precedent disease: Hemophilia A
Vector: AAV8
Tissue target: liver
Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 1599bp / cargo 4700bp (34% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: phase2

Match #5: DTX301

Precedent disease: Ornithine transcarbamylase deficiency
Vector: AAV8
Tissue target: liver
Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 1599bp / cargo 4700bp (34% utilized)

- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: phase2